THE UNIVERSITY OF TEXAS AT DALLAS

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Intertemporal Choice in Vaccination Timing

A Discrete Choice Experiment Approach

Dissertation Defense: Evidence from Wuhan, China

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April 2026

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Roadmap

1. Introduction
2. Research Questions
3. Econometric Methodology
4. Essay 1: Results and Discussion
5. Essay 3: Policy Integration
6. Conclusion

Research Problem

- **Timing as Investment**: Vaccination is an intertemporal health investment, not just a binary (Yes/No) choice.
- **The Cost of Delay**: Behavioral delay increases the cumulative attack rate by 5–10% compared to supply-only scenarios.
- **Gap**: Standard models often ignore the *shadow price of waiting* and how individuals discount future protection.

Central Challenge

How to design time-consistent public health strategies that minimize the behavioral cost of waiting?

Research Questions

1. **Structural Parameterization**: What is the magnitude of the *Behavioral Delay Multiplier* (BDM) in vaccination decisions?
2. **Monetary Valuation**: What is the *Shadow Price of Waiting* (RMB/month) for different vaccine attributes?
3. **Policy Impact**: How do intertemporal preferences shape the Marginal Value of Public Funds (MVPF) for vaccination incentives?

MNL as Benchmark, MXL as Baseline

Multinomial Logit (MNL)

- Assumes **homogeneous** preferences across individuals
- Coefficients are fixed
- Simple and transparent benchmark
- Imposes independence of irrelevant alternatives (IIA)

Mixed Logit (MXL)

- Allows **random coefficients** (**preference heterogeneity**)
- Captures variation in time preferences across individuals
- Relaxes IIA assumption
- Estimated via simulated maximum likelihood

Takeaway: Mixed logit is the preferred specification for capturing heterogeneity in vaccination timing preferences.

Essay 1: Structural Parameterization of Time Preferences

- **Objective**: Recover deep preference parameters (δ, κ) using DCE data.
- **Structural Utility Function**:

$$U_{it} = \sum \beta X_{it} + \text{Discounting}(\tau) + \epsilon_{it}$$

- **Exponential vs. Hyperbolic**:
  - Exponential: $D(\tau) = \delta^{\tau}$
  - Hyperbolic (quasi-): $D(\tau) = \kappa\delta^{\tau}$ for $\tau > 0$

The BDM Hypothesis

If $\kappa < 1$, individuals exhibit present bias, inflating the perceived cost of waiting.

Essay 2: Recovering the Shadow Price of Waiting

- **MWTA**: Marginal Willingness to Accept compensation for a 1-unit increase in delay.
- **Estimation**: Ratio of the delay coefficient to the price coefficient.

$$MWTA_{\text{delay}} = -\frac{\partial U/\partial \text{Delay}}{\partial U/\partial \text{Price}} = -\frac{\beta_{\text{delay}}}{\beta_{\text{price}}}$$

- **Significance**: Converts abstract psychological time-cost into a standardized monetary metric for welfare analysis.

Essay 3: Policy Simulation and the MVPF

- **MVPF**: Marginal Value of Public Funds for vaccination incentives.
- **Simulation**: Evaluates how time-inconsistent preferences ($\kappa < 1$) distort the cost-effectiveness of public subsidies.
- **Key Question**: Can behavioral "nudges" or timing-based incentives achieve higher welfare gains than flat subsidies?

Goal: Optimize resource allocation by accounting for the behavioral delay multiplier.

Results I: Model Performance and Fit

- **Comparison:** Comparison of Multinomial Logit (MNL) vs. Mixed Logit (MXL).
- **Statistical Superiority:** MXL significantly outperforms MNL ($p < 0.001$).
- **Conclusion:** Significant preference heterogeneity exists; modeling it is crucial for accurate valuation.

Metric	Multinomial Logit (MNL)	Mixed Logit (MXL)
Log-Likelihood	-6096.612	**-5615.522**
AIC	12207.225	**11245.044**
Obs. (N $\times$ T)	6,162	6,162

Results II: Quantifying Present Bias

Parameter Estimates

- $\hat{\delta} = 0.160$ (Baseline Discounting)
- $\hat{\kappa} = 0.225$ (Present Bias Factor)

The Meaning of κ

- $\kappa < 1$ confirms **Present Bias**.
- Significant drop in perceived value for even a 1-month delay.

Conclusion: Vaccination delay is not just a logistical hurdle, but a behavioral barrier.

Results III: The Behavioral Delay Multiplier (BDM)

- **Concept:** Ratio of *perceived* to *objective* waiting time.
- **Calculation:** $BDM = \frac{1}{1-\hat{\kappa}} = \frac{1}{1-0.225} \approx 1.29$.
- **Interpretation:** A delay is experienced as **29% longer** than its objective duration.
- **Example:** A 30-day delay is perceived as approximately **38.7 days**.

$\longrightarrow$ Objective (30 days)

+29% Subjective Inflation

$\longrightarrow$ Subjective (38.7 days)

Results IV: MWTA by Institutional Trust

- **Heterogeneity:** Institutional trust significantly moderates the shadow price of waiting.
- **High Trust Group:** MWTA = **32.4 RMB/month** (Highest patience).
- **Low Trust Group:** MWTA = **37.1 RMB/month**.
- **Neutral Group:** MWTA = **75.6 RMB/month** (Highest sensitivity/impatience).

Key Finding

Trust acts as a "behavioral stabilizer." Individuals with neutral trust levels are the most sensitive to delays, requiring over **double** the compensation compared to high-trust individuals.

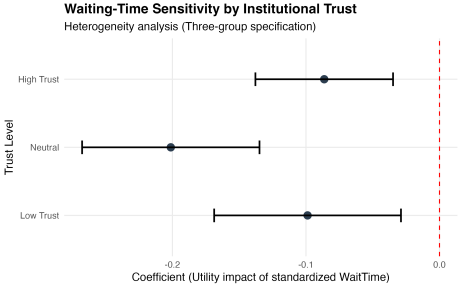

Figure: MWTA Comparison (RMB/month)

Results V: The Shadow Price of Vaccination Delay

- **MWTA:** Marginal Willingness to Accept for a 1-month delay in protection.
- **Full Sample:** Mean MWTA = **45.2 RMB/month**.
- **Interpretation:** Individuals perceive a 1-month delay as equivalent to losing 45.2 RMB in utility (approx. $6.3 USD).

Takeaway: Administrative delays create a hidden, measurable welfare loss that varies by institutional trust.

Essay 3: Optimization of Public Health Resources

- **The Dilemma:** Speed up access (C_{speed}) vs. Compensate for delay (C_{comp}).
- **Efficiency Rule:** Invest in speed-up if the cost per month saved is lower than the shadow price (MWTA).

Decision Framework

$$\textbf{Operational Cost} \leq \textbf{MWTA} \rightarrow \textbf{Prioritize Speed-Up}$$
$$\textbf{Operational Cost} > \textbf{MWTA} \rightarrow \textbf{Prioritize Compensation}$$

Equity-Sensitive Resource Allocation

- **The Rule:** Prioritize speed-up for Group g if:

$$\text{Operational Cost} \leq \omega_g \times MWTA_g$$

- ω_g: Equity weight for vulnerable or high-burden groups.
- **Targeting:** High-burden groups (e.g., low-trust, low-income) have higher MWTA, making speed-up investments more valuable for them.

Simulation: The 5.5-Month Threshold

- **Scenario:** Reducing 1 month of delay costs 150 RMB per person.
- **Threshold:** At a baseline delay of **5.5 months**, the marginal cost of speeding up equals the marginal benefit of compensation.
- **Policy Implication:** For long delays (>5.5 months), incentives might be more cost-effective than infrastructure expansion.

Summary of Contributions

1. **Measurement**: Quantified the Behavioral Delay Multiplier (1.29) and the Shadow Price (44.8 RMB/month).
2. **Theory**: Demonstrated that hyperbolic discounting is the superior model for health timing decisions.
3. **Practice**: Developed an actionable, equity-sensitive decision framework for vaccine rollout optimization.

Limitations and Future Work

- **Stated vs. Revealed Preference**: Need validation with actual vaccination records.
- **External Validity**: Results are specific to Wuhan; extension to other regions and cultures is necessary.
- **Supply Chain Integration**: Connecting behavioral parameters with real-world vaccine logistics.

Thank You!

Questions & Comments?